

EXPANCIFY

A Mini Project Report

Submitted in partial fulfillment of requirement of the

Degree of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY

BY

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Report Approval

The project work “**Expancify**” is hereby approved as a creditable study of an engineering/computer application subject carried out and presented in a manner satisfactory to warrant its acceptance as prerequisite for the Degree for which it has been submitted.

It is to be understood that by this approval the undersigned do not endorse or approved any statement made, opinion expressed, or conclusion drawn there in; but approve the Mini Project Report only for the purpose for which it has been submitted.

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Declaration

I/We hereby declare that the project entitled “**Expancify**” submitted in partial fulfillment for the award of the degree of Bachelor of Technology in ‘Information Technology’ completed under the supervision of Dinesh OKumar Bhayal, Assistant Professor, Department of Information Technology , Faculty of Engineering, Medicaps University Indore is an authentic work.

Further, I/we declare that the content of this Mini Project work, in full or in parts, have neither been taken from any other source nor have been submitted to any other Institute or University for the award of any degree or diploma.

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Certificate

I, Dinesh Bhayal certify that the mini project entitled “Expansify” submitted in partial fulfillment for the award of the degree of Bachelor of Technology by Kunal Choudhary is the record carried out by him/them under my/our guidance and that the work has not formed the basis of award of any other degree elsewhere.

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Abstract

Expancify is an AI-powered expense tracking web application that helps users manage their daily spending in a simple and effective way. The main goal of this project is to make users more aware of their financial habits by allowing them to manually add and track their expenses.

A key feature of the system is its AI-based analysis, which identifies unnecessary or unwanted expenses. This helps users understand where they are overspending and guides them towards better financial decisions.

The application also supports shared expense tracking, where users can manage and split expenses with their roommates. This makes it easier to handle common costs like rent, groceries, and other shared payments.

Overall, Expancify provides a user-friendly platform for both personal and group expense management, while also promoting smarter spending through AI-driven insights.

Keywords:

Expense Tracking, AI-powered Analysis, Unwanted Expense Detection, Personal Finance Management, Shared Expense Tracking, Roommate Expense Management, Web Application

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Abbreviations

- AI – Artificial Intelligence
- UI – User Interface
- API – Application Programming Interface
- DB – Database
- SRS – Software Requirements Specification
- IT – Information Technology

Chapter-1: INTRODUCTION

1.1 Background of the Study

Managing personal expenses has become an important part of daily life, especially for students and working individuals. Many people find it difficult to keep track of their daily spending, which often leads to unnecessary expenses. In most cases, people either forget to record their expenses or do not maintain them properly. Traditional methods like writing expenses in notebooks or using simple spreadsheets are time-consuming and not very effective. Therefore, there is a need for a simple and user-friendly system that can help users manage their expenses in a better way.

1.2 Problem Statement

Many existing methods of expense tracking are not efficient and do not provide proper insights into spending habits. Users are often unaware of where their money is being spent, which results in poor financial decisions. It is also difficult to identify unnecessary expenses without proper analysis. In addition, managing shared expenses with roommates becomes confusing due to lack of transparency and proper tracking. Hence, there is a need for a system that can track expenses, analyze spending patterns, and manage shared expenses easily.

1.3 Objectives of the Project

The main objectives of the Expancify project are:

- To develop a system for tracking daily expenses
- To allow users to manually add, update, and delete their expenses
- To identify unnecessary or unwanted expenses using AI-based analysis
- To provide a feature for managing and splitting expenses with roommates
- To improve financial awareness and spending habits of users

1.4 Scope of the Project

The scope of Expancify is limited to personal and shared expense management. The system is designed mainly for students and individuals who want to keep track of their daily spending. It provides features like expense tracking, AI-based analysis, and roommate expense management. However, the system does not include features like direct bank integration or real-time financial transactions. The application focuses only on helping users manage and analyze their expenses effectively.

1.5 Overview of the System

Expancify is a web-based application that helps users manage their expenses in a simple and organized way. Users can manually add their expenses and keep track of their spending. The system also uses AI-based analysis to identify unnecessary expenses and provide useful insights. Additionally, it allows users to manage shared expenses with their roommates, making it easier to handle common costs like rent, food, and utilities. Overall, the system aims to provide a simple and effective solution for personal and group expense management.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Literature review is an important part of any project as it helps to understand the existing systems and technologies related to the problem. It provides an overview of the methods and applications that are already being used for expense management. By studying these systems, it becomes easier to identify their limitations and find ways to improve them.

2.2 Existing Expense Tracking Systems

There are many expense tracking applications available today that help users manage their daily expenses. These applications allow users to record their spending, categorize expenses, and view summaries. Some applications also provide graphical representations like charts and reports to help users understand their spending patterns.

However, most of these systems focus only on recording expenses and do not provide deep insights into user behavior. They require users to manually analyze their spending, which may not be convenient for everyone.

2.3 Limitations of Existing Systems

Although existing expense tracking systems are useful, they have several limitations:

- They do not effectively identify unnecessary or unwanted expenses
- Most systems lack intelligent analysis of spending habits
- Users need to manually understand their financial patterns
- Shared expense tracking is often complex or not user-friendly
- Many applications have complicated interfaces which are difficult for beginners

These limitations make it difficult for users to fully understand and control their spending.

2.4 Need for the Proposed System

Due to the limitations of existing systems, there is a need for a more advanced and user-friendly solution. Users require a system that not only tracks expenses but also helps them understand their spending behavior. There is also a need for better shared expense management, especially for students and individuals living with roommates.

An improved system should provide intelligent analysis, simple user interface, and easy management of both personal and shared expenses.

2.5 Proposed System Overview

To overcome the limitations of existing systems, the project “Expancify” is proposed. It is an AI-powered expense tracking system that allows users to manage their expenses efficiently. The system provides features like manual expense entry, tracking, and AI-based analysis to identify unnecessary expenses.

In addition, Expancify offers a shared expense management feature, which helps users track and split expenses with their roommates in a simple and transparent way. The system focuses on providing a clean and user-friendly interface, making it easy for users to manage their finances.

CHAPTER 3: PROPOSED SYSTEM

3.1 Introduction

The proposed system, Expancify, is designed to provide a simple and efficient way for users to manage their expenses. It focuses on helping users track their daily spending, analyze their financial habits, and manage shared expenses with roommates. The system is developed as a web-based application with a user-friendly interface.

3.2 System Overview

Expancify allows users to manually add, update, and delete their expenses. All expenses are stored and organized in a structured format so that users can easily view and manage them. The system also includes an AI-based analysis feature that helps identify unnecessary expenses and provides useful insights.

In addition, the system supports shared expense tracking, where users can manage and split expenses with their roommates. This feature makes it easier to handle common costs like rent, groceries, and utilities.

3.3 Functional Requirements

The system provides the following functionalities:

- User registration and login
- Adding new expenses
- Editing and deleting existing expenses
- Viewing and managing expense records
- AI-based analysis for identifying unwanted expenses
- Shared expense tracking with roommates
- Secure user authentication

3.4 Non-Functional Requirements

The system should satisfy the following non-functional requirements:

- The application should be user-friendly and easy to use
- The system should provide fast response time
- Data should be stored securely
- The system should be accessible through web browsers
- The application should handle multiple users efficiently

3.5 Advantages of Proposed System

The proposed system offers several advantages:

- Easy and simple expense tracking
- Helps identify unnecessary spending using AI
- Supports both personal and shared expense management
- User-friendly interface for better experience
- Improves financial awareness and decision-making

CHAPTER 4: SYSTEM DESIGN

4.1 Introduction

The system design of Expancify explains the overall structure and working of the application. It shows how different components interact with each other and how data flows within the system. Various diagrams are used in this chapter to provide a clear understanding of the system.

4.2 System Architecture Diagram

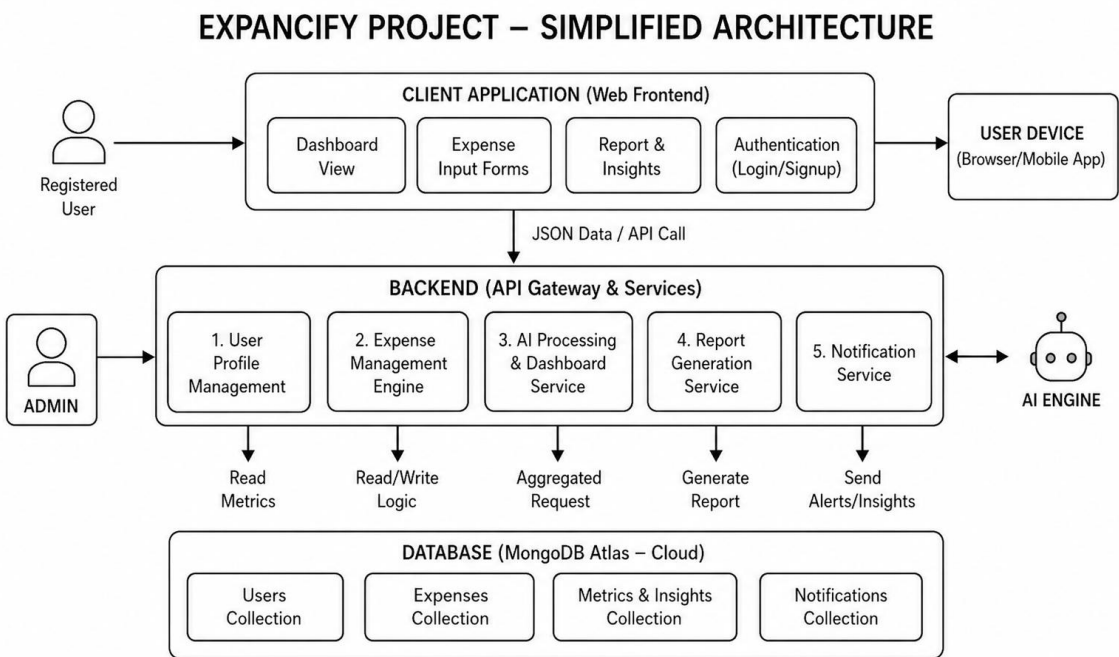


Figure 4.2: System Architecture Diagram

The system architecture shows the overall structure of Expancify. It consists of the user interface, backend server, database, and AI module. The user interacts with the frontend, which sends requests to the backend. The backend processes the data, communicates with the database, and returns the response to the user. The AI module analyses expense data to identify unnecessary expenses.

4.3 Use Case Diagram

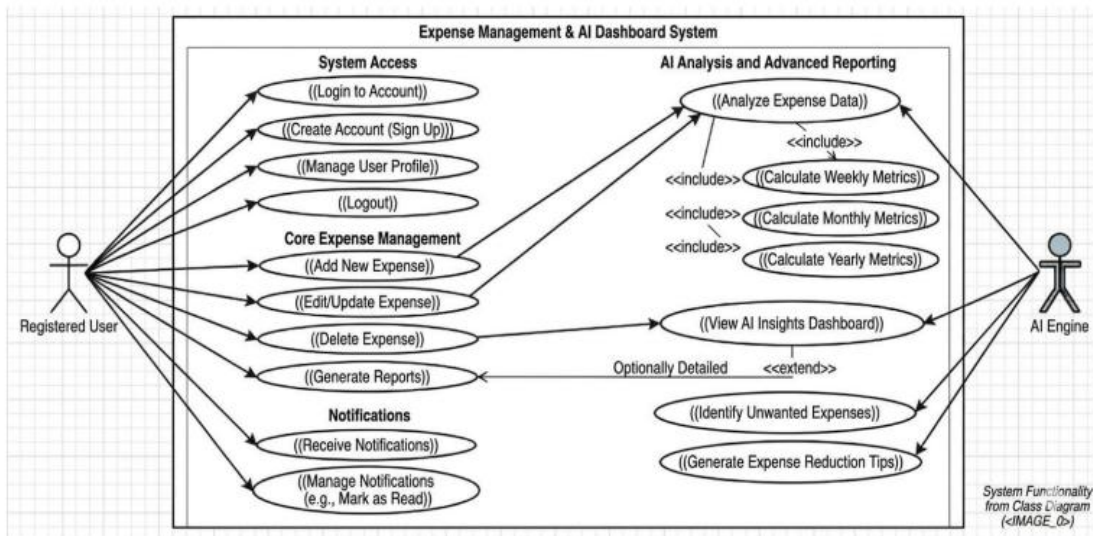
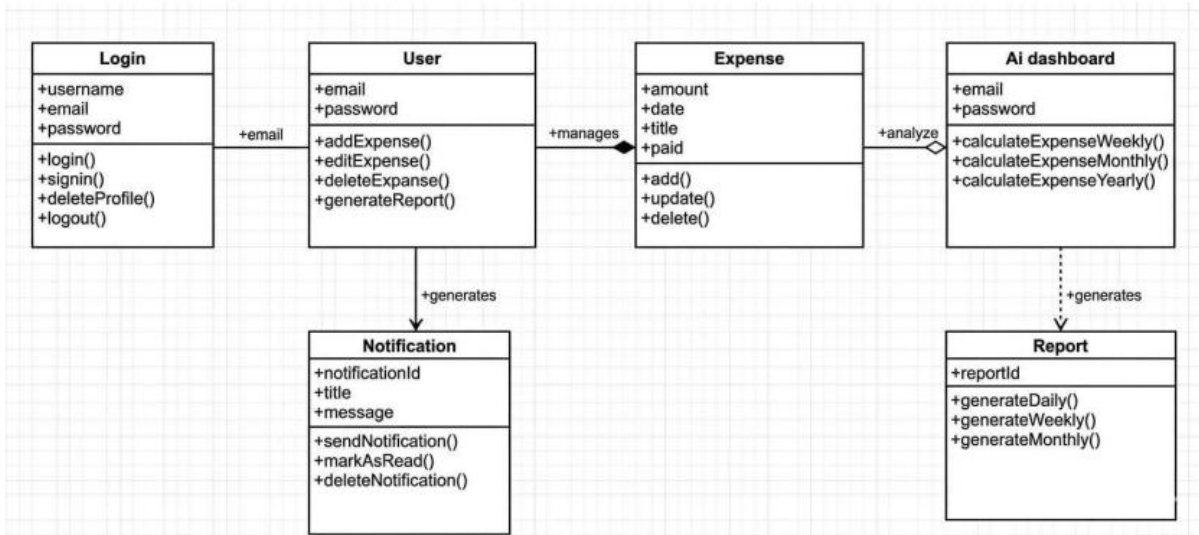


Figure 4.3: Use Case Diagram

The use case diagram represents the interaction between the user and the system. The user can perform actions such as registration, login, adding expenses, viewing expenses, editing or deleting expenses, analyzing expenses, and managing shared expenses with roommates.

4.4 Class Diagram



Class Diagram of Expansify

Figure 4.4: Class Diagram

The class diagram shows the structure of the system in terms of classes and their relationships. It includes classes such as User, Expense, and Roommate. These classes define the properties and functions required for managing expense data and user information.

4.5 Activity Diagram

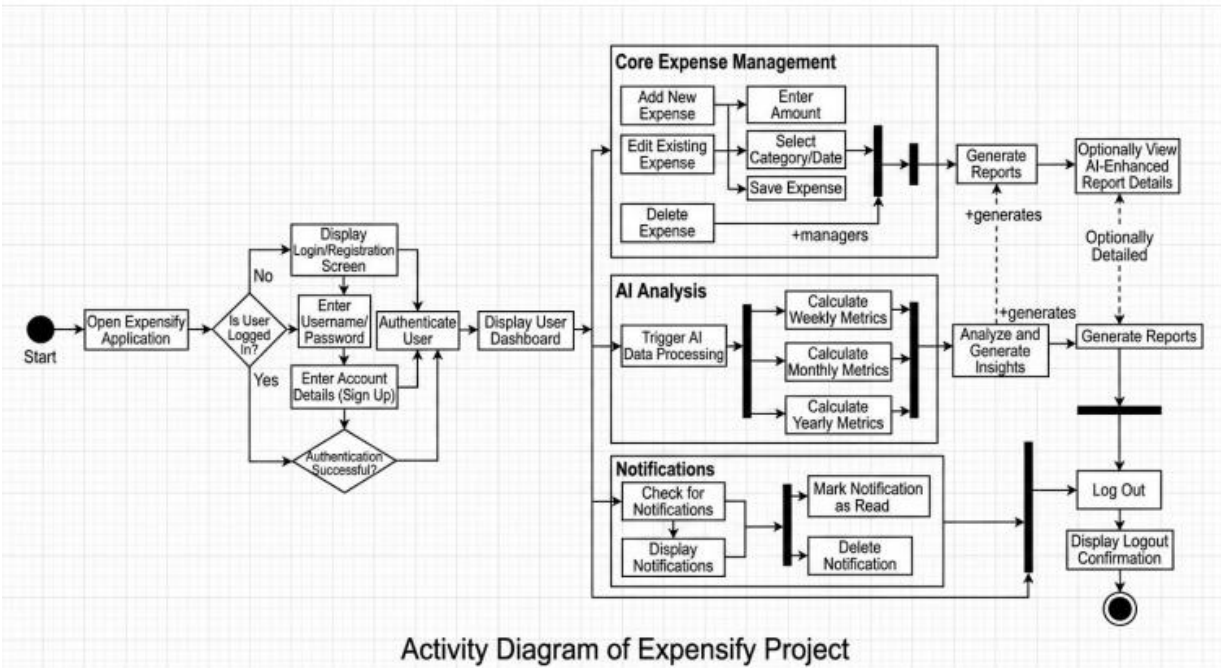


Figure 4.5: Activity Diagram

The activity diagram represents the workflow of the system. It shows the sequence of activities such as user login, adding expenses, processing data, and displaying results. It helps in understanding how the system operates step by step.

4.6 ER Diagram

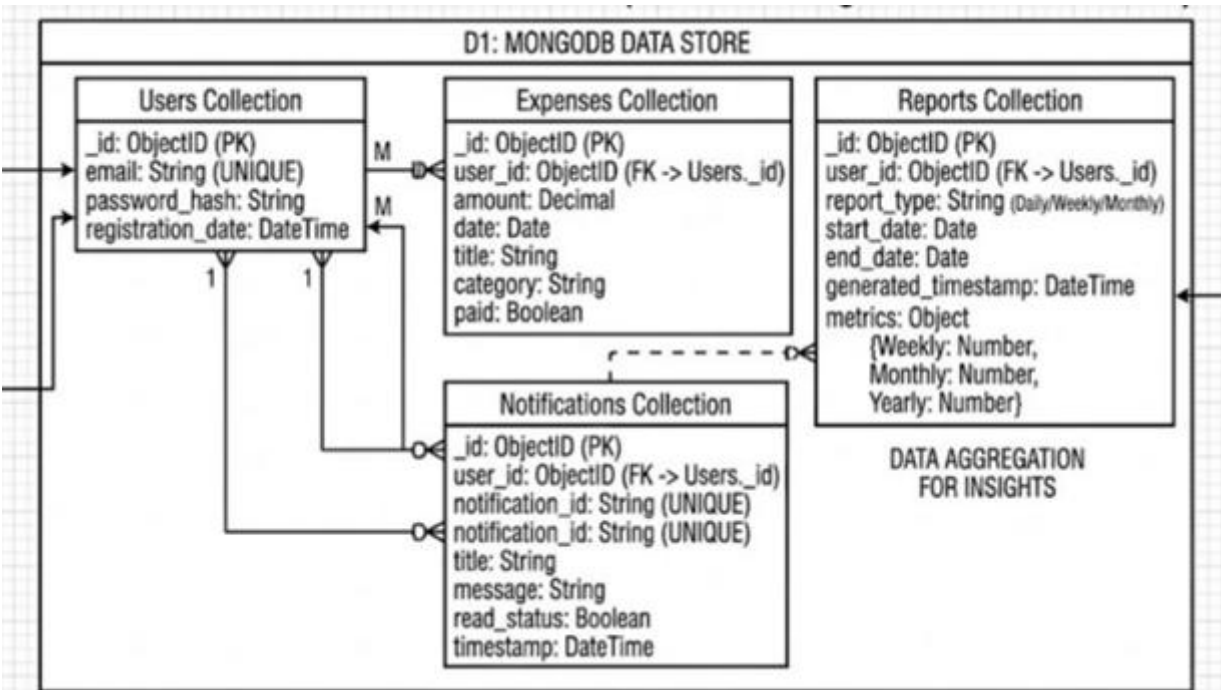


Figure 4.6: ER Diagram

The ER diagram represents the database structure of the system. It shows entities such as User, Expense, and Roommate along with their relationships. This diagram helps in understanding how data is stored and connected in the database.

4.7 System Flow Diagram / Data Flow Diagram

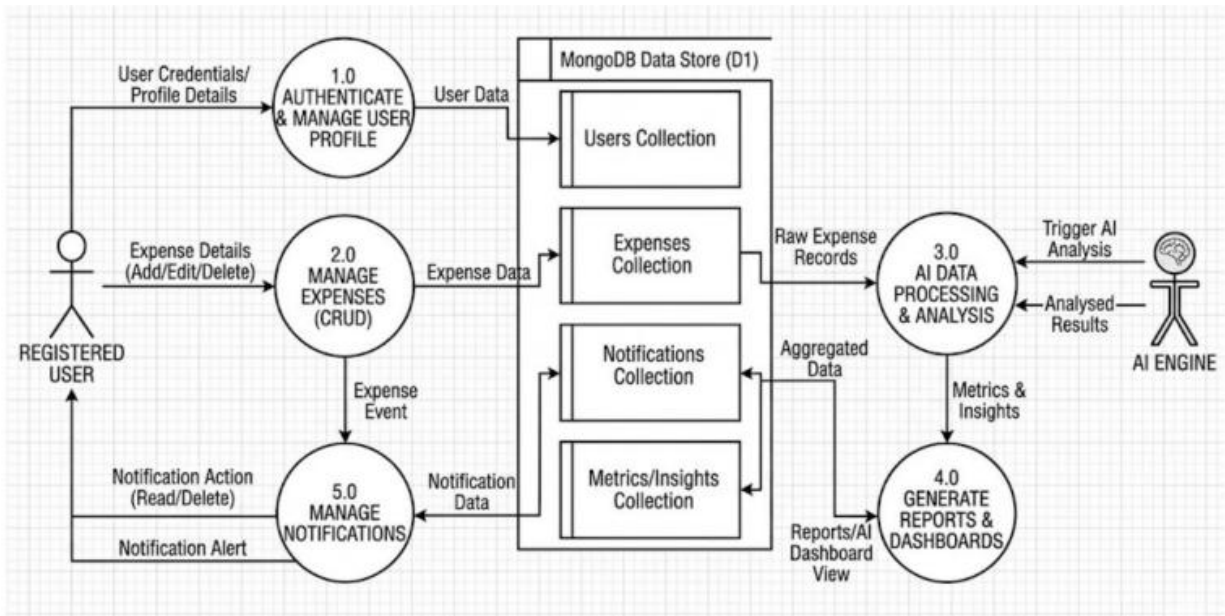


Figure 4.7: Data Flow Diagram

The data flow diagram shows how data moves within the system. The user inputs expense data, which is processed by the system and stored in the database. The system retrieves and displays the data when required. The AI module also uses this data for analysis.

4.8 Sequence Diagram

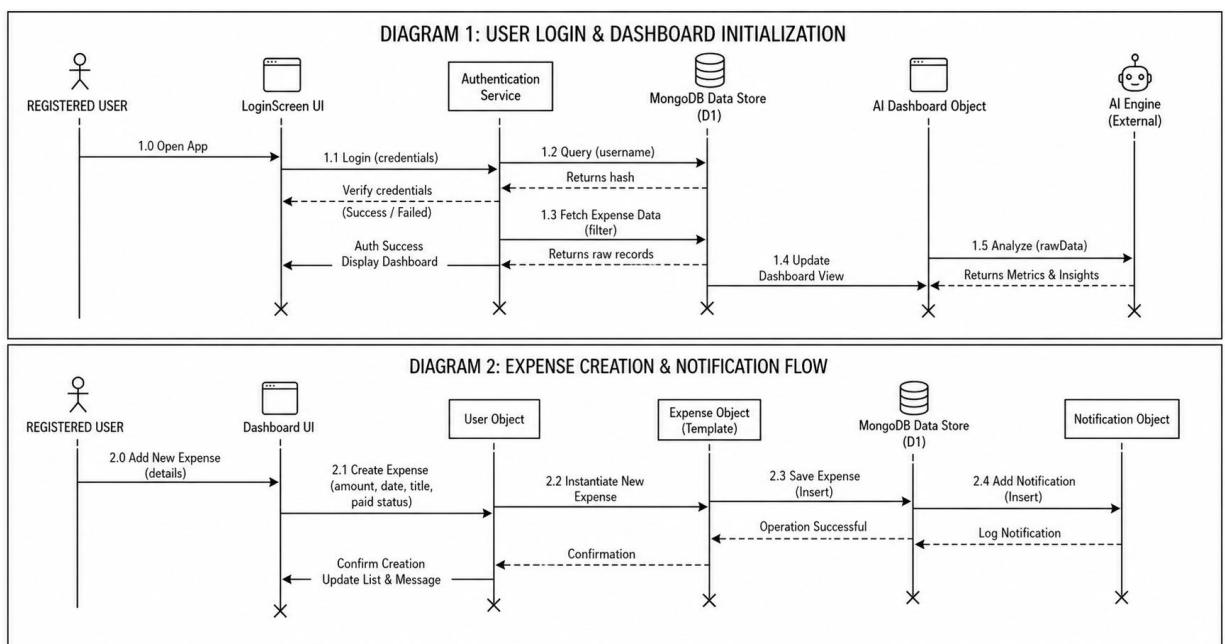


Figure 4.8: Sequence Diagram

The sequence diagram shows the interaction between different components of the system in a time sequence. It describes how requests are sent from the user to the system and how responses are returned. It helps in understanding the step-by-step execution of system operations.

CHAPTER 5: SUMMARY AND CONCLUSION

5.1 Summary

The Expancify project is developed as an AI-powered expense tracking system that helps users manage their daily expenses in an efficient and organized manner. The system allows users to manually add, update, and delete their expenses, making it easy to keep track of spending.

The application also provides an AI-based analysis feature, which helps in identifying unnecessary or unwanted expenses. This feature helps users understand their spending patterns and improve their financial habits.

In addition, the system supports shared expense management, allowing users to track and split expenses with their roommates. This makes it easier to manage shared costs like rent, food, and utilities.

Overall, the system provides a simple and user-friendly solution for both personal and group expense management.

5.2 Conclusion

The Expancify system successfully achieves its main objective of helping users manage their expenses effectively. The system is easy to use and provides useful insights into spending behavior. The integration of AI-based analysis makes the system more intelligent compared to traditional expense tracking methods.

The shared expense feature also improves usability for users living with roommates. The system can be considered a useful tool for improving financial awareness and decision-making.

5.3 Future Scope

Although the system works effectively, there are several improvements that can be made in the future:

- Integration with bank accounts for automatic expense tracking
- Advanced AI models for better analysis and predictions
- Mobile application development
- Adding budget planning and notifications
- Improved data visualization using charts and graphs

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